

Random Practice Can Facilitate the Learning of Tasks That Have Different Relative Time Structures

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The contextual interference (CI) effect has been replicated many times since its first demonstration by Shea and Morgan (1979) in the motor learning domain (see Brady, 1998; Magill & Hall, 1990). The CI effect is characterized by the observation that experiencing greater interference during acquisition is detrimental to immediate performance but enhances delayed performance as measured on retention or transfer tests. High CI is most often created by random practice in which the learning of multiple tasks occurs in a single training. In contrast, low CI is frequently created by using a blocked practice format in which all the practice trials of one task are completed before another task is introduced. One theoretical account that has been forwarded to account for the CI effect is labeled the *action plan reconstruction* hypothesis (Lee & Magill, 1983, 1985). This position intimates that before a movement occurs an “action plan” must be prepared. In blocked practice, a previously prepared “action plan” is readily available from trial to trial, but it suffers from lack of attention on trials following initial retrieval from working memory. In random practice, however, each time a task must be executed a “reconstruction” of the action plan must be processed, because the interchange of information from trial to trial never allows the same information to remain in working memory for an extended amount of time.

Presumably, the additional trial-to-trial preparation used by the random practice participant during practice results in a more resilient memory representation that better supports long-term recall efforts compared to their blocked practice counterparts.

One issue that emerges from the action plan reconstruction account, but has only received cursory attention to date, is the nature or extent of reconstructive activity that must occur in working memory for random practice to afford the aforementioned learning benefits. An initial attempt to address this issue by Magill and Hall (1990) used motor program theory¹ (Schmidt, 1975, 1985) as a theoretical backdrop and considered the impact of the trial-to-trial interference created by required changes in the generalized motor program (GMP) or parameters. Specifically, Magill and Hall (1990) proposed that when task variations are governed by different GMPs, random practice would lead to better retention and transfer performance than blocked practice, presumably because the nature of this interference would be sufficient to encourage reconstructive activity prior to the next trial. This is consistent with the notion that structural characteristics (i.e., invariant characteristics such as relative time) must be rectified prior to resolving the absolute characteristics of the movement, such as absolute time or force (see Wright, Black, Park, & Shea, 2001; Shea, Lai, Wright, Immink, & Black, 2001). Magill and Hall also suggested that when tasks were governed by the same GMP (thus, only requiring the movement be scaled appropriately), the CI effect would not be present. This implies that parameter modification from trial-to-trial (i.e., absolute time and force), even in high CI conditions such as random practice, do not create enough interference to warrant action plan reconstruction.

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Magill and Hall's (1990) claim that dealing with tasks only requiring scaling adjustments during practice would not result in robust CI effects appears to be incorrect, given evidence to the contrary (see Shea, Kohl, & Indermill, 1990; Sekiya, Magill, & Anderson, 1996). However, there appears to be some support for Magill and Hall's (1990) initial assertion that the CI effect will surface when practicing tasks governed by different GMPs. This evidence is based primarily on studies in which the tasks being acquired were assumed to use different GMPs because of the tasks' unique spatial layouts. For example, such data are apparent from early work using a barrier knockdown task (e.g., Lee & Magill, 1983; Shea & Morgan, 1979) or more applied tasks such as badminton serves (Goode & Magill, 1986). However, evidence from studies involving commonly used a priori manipulations of tasks characteristics specific to motor program theory (e.g., relative time structure; for recent examples, see Lai & Shea, 1998, 1999) is sparse in supporting Magill and Hall's (1990) claim.

A study by Sekiya, Magill, Sidaway, and Anderson (1994), involving the learning of movements, differed with respect to predetermined relative time structures. Their results revealed that individuals assigned to a high CI practice condition resulted in greater parameter learning but similar GMP learning when compared to their blocked practice counterparts. These findings are consistent with previous studies demonstrating that high CI training conditions result in superior learning of overall duration of multisegment tasks (i.e. Shea and Morgan, 1979) and appear, at least in part, congruent with Magill and Hall's (1990) claim. However, before accepting these data as support for their position, one should note that overall duration was also manipulated in the Sekiya et al. (1994) study. Thus, one cannot be sure if the interference introduced into practice *merely* from handling changes in the GMP is sufficient to support random practice benefits.

Thus, the main purpose of the present investigation was to test directly the initial claim of Magill and Hall (1990) that practicing tasks governed by different GMPs create enough interference to engender a learning benefit from random practice. To address this issue, we used three three-segment key pressing sequences that had predetermined relative and absolute timing structures. Specifically, participants practiced sequences with different relative timing structures *but shared* the same overall duration. Emergence of the CI effect in this study would support Magill and Hall's (1990) contention. In addition to the aforementioned primary goal, we also wanted to probe the generalizability of any learning that might be observed from random or blocked practice of movement sequences characterized only by different relative time structures. For example, it has been demonstrated on numerous occasions that ran-

dom practice is *particularly effective* for engendering transfer to new absolute scaling requirement. However, it has never, at least to our knowledge, been determined whether the scope of such generalizability from high CI training is broader than currently documented. Thus, the present experiment included a transfer test composed of a movement sequence that had a relative time structure different from the trained sequences.

Method

Participants

Twenty-four right-handed undergraduate students enrolled in kinesiology classes were recruited to participate in the experiment for course credit. All participants read and signed an informed consent form before participating in the experiment.

Apparatus and Tasks

A standard computer, color monitor, and keyboard were positioned on a table in an enclosed testing room. Participants were asked to sit in a comfortable chair so that they could easily access the keyboard with their right hand. All experimental procedures were programmed with E-Prime® Beta version 5.0 software. The keys on the numeric keypad portion of a standard keyboard were covered with colored tape to eliminate the use of the numbers associated with specific keys.

During the acquisition and retention phases of the experiment, each participant was required to perform three sequential key pressing tasks. Each task required them to depress four keys (# 2, 4, 8, and 6) on the numeric pad portion of the computer keyboard using the index finger of their right hand. Prior to each trial, the relative goal proportions for each movement segment (with segments defined by two consecutive key presses) and the total goal movement time (GMT) were presented on the computer monitor. The tasks differed with respect to their relative time requirement but had the same GMT. The GMT for all the tasks was 1,150 ms. The relative goal segment ratios were 33.3, 44.4, and 22.2%; 22.2, 33.3, and 44.4; and 44.4, 22.2, and 33.3% for all groups. For the transfer test, the task differed only in terms of the relative time structure, which was 39, 25, and 36%.

Design

An equal number of participants ($N=12$) were randomly assigned to a random or blocked practice condi-

tion. All participants experienced three blocks of 45 trials. In the blocked condition, all trials within a block involved only one task. Thus, individuals in blocked practice were administered all 45 trials of one task before receiving any practice on one of the alternative tasks. Participants in the random condition were administered practice trials on each of the tasks in a quasirandom order, with the stipulation that 15 trials of each task were included in each block of 45 trials. After completing the acquisition trials, all participants performed retention and transfer tests 24 hr after the completion of acquisition. They completed the retention test, which consisted of 15 trials (five of each task) presented in a random order, first. The transfer test also consisted of 15 trials but of a novel task, which had the same GMT (1,150 ms) but different goal segment ratios (39, 25, and 36%) than the tasks assessed during the retention test.

Procedure

Participants were tested individually in a room without distractions. The participants were instructed that the objective of the experiment was to depress specific keys on the numeric keypad of the keyboard in the prescribed order, using the index finger of their right hand. They were instructed to be as accurate as possible with respect to the instructed relative goal segment ratios and the GMT. These instructions were presented prior to acquisition, retention, and transfer trials.

Before each trial, the words "get ready" appeared on the computer monitor. Following a variable fore period, the imperative stimulus in the form, "Task A 33.3% 44.4% 22.2% 1150," was presented in the middle of the computer display. At the onset of the stimulus, the participant was instructed to execute this task. If the participant depressed the keys in the wrong order, an error was recorded and the trial was repeated. Knowledge of results (KR) concerning the participant's goal segment ratios and total movement time was displayed under the respective goal segment ratio times and GMT immediately following a correct response for a duration of 8 s. A 1-min rest interval was provided between each block of 45 trials during acquisition. After the completion of all acquisition trials, 24-hr delayed retention and transfer tests were administered. The same procedure was followed for the retention and transfer tests, with the exception that KR for both the relative and absolute times were not provided.

Treatment and Analysis of Data

The dependent variables of interest were relative timing error (RTE) and absolute timing error (E). RTE was used as a measure of the proficiency of the GMP, and

E was used as a measure of the proficiency of the parameterization (Lai & Shea, 1998). RTE was computed as the sum of the absolute differences between the goal segment ratios and the actual proportions performed on each segment. This measure provides an estimate of the accuracy of relative timing performance. Thus, the calculation was:

$$\text{Relative timing error (RTE)} = |R_1 - S_1| + |R_2 - S_2| + |R_3 - S_3|$$

where R_n = (the actual MT/ total MT) and S_n = (relative goal segment ratios). Thus, R_1 , R_2 , and R_3 were the proportion of the total movement time used in segments 1, 2, and 3, respectively. Whereas, S_1 , S_2 , S_3 were the relative goal segment ratios for segments 1, 2, and 3, respectively.

For absolute timing performance, we used E to characterize parameter errors. This measure is one of overall accuracy in responding that considers both response biases and response variability. Thus, absolute timing error was computed as:

$$\text{Absolute timing error (E)} = (CE^2 + VE^2)^{1/2}$$

where CE is a measure of absolute timing bias and computed as the average of the signed errors (GMT - MT). VE is a measure of the variability in absolute timing and computed as the standard deviation of the signed errors.

Results

In acquisition, mean RTE and E measures were calculated for each individual for nine blocks of 15 trials. For blocked practice, Block 1 consisted of the first five trials of each task; Block 2 consisted of the second five trials of each task, and so on. For random practice, Block 1 consisted of the first 15 trials (5 of each task), Block 2 consisted of the second 15 trials (5 of each task), and so on. The RTE and E data were subsequently submitted to separate 2 (practice condition: block/random) $\times$ 9 (block: 1-9) analyses of variance (ANOVAs) with repeated measures on the last factor.

For retention, mean RTE and E measures were averaged into one block of 15 trials. These measures were evaluated with separate one-way between-factor (practice condition: blocked/random) ANOVAs. The RTE and E data from the transfer test were also submitted to separate one-way between-factor (practice condition: blocked/random) ANOVAs. In all analyses, the rejection region was $p < .05$. The Greenhouse-Geisser adjusted F test was used in computing significance for all within-participant analyses in which the sphericity assumption was violated.

Acquisition

Relative Timing Error. Figure 1 displays mean RTE (top panel) and E (top panel) blocked and random practice acquisition conditions for the acquisition, retention, and transfer blocks. The analysis of mean RTE revealed significant main effects of practice condition $F(1, 22) = 5.91, p < .05$, and trial block, $F(5, 102) = 14.41, p < .01$. Mean RTE for the individuals in the random practice condition ($M = 0.23, S_x = .01$) was significantly higher than mean RTE for their blocked practice counterparts ($M = .18, S_x = .01$) during acquisition. Post hoc analyses of the trial block main effect indicated that mean RTE for Block 1 was higher than mean RTE exhibited for Block 2, which in turn was higher than the mean RTE observed for Blocks 3–9. No interactive effects were reliable.

Absolute Timing Error. The analysis of mean E revealed a significant main effect of block, $F(3, 62) = 3.01, p < .05$. The post hoc analysis revealed that mean E for Block 1 was significantly lower than mean E for all other blocks during acquisition. Mean E for Blocks 2–9 were not reliably different. No other main effects or interactive effects were significant.

Retention

Relative Timing Error. The analysis of mean RTE for the retention block indicated a significant main effect of practice condition $F(1, 22) = 8.72, p < .01$. The main effect was a result of lower mean RTE for the individuals assigned to random practice condition ($M = .19, S_x = .02$) compared to that observed for the individuals who experienced blocked practice ($M = .25, S_x = .01$).

Absolute Timing Error. The analysis of mean E revealed a significant main effect of practice condition, $F(1, 22) = 6.50, p < .05$. This main effect was a function of lower mean E for the participants in the random practice condition ($M = 233 \text{ ms}, S_x = 33$) compared to those assigned to blocked practice ($M = 392 \text{ ms}, S_x = 183$).

Transfer

Relative Timing Error. As was the case for retention, the analysis of mean RTE for the transfer block revealed a significant main effect of practice condition, $F(1, 22) = 10.48, p < .01$. A lower mean RTE was observed for the individuals that experienced random practice ($M = .18, S_x = .05$) as opposed to blocked practice ($M = .24, S_x = .05$) during acquisition.

Absolute Timing Error. Consistent with analysis of the retention data, the analysis of mean E for the transfer block revealed a significant main effect of practice condition, $F(1, 22) = 4.31, p < .05$. This main effect was a result of mean E being lower for the participants that experienced random practice ($M = 222 \text{ ms}, S_x = 30$) as opposed to blocked practice during acquisition ($M = 399 \text{ ms}, S_x = 80$).

Discussion

The primary purpose of the present study was to test the claim of Magill and Hall (1990) that practicing tasks governed by a different GMP creates enough interference to engender a learning benefit from random practice that is greater than that observed from blocked practice. While previous studies have provided some support for Magill and Hall's suggestion, the support is from studies that assumed the tasks differed with respect to the GMPs governing their execution because of their unique spatial arrangements (for an example, see Shea & Morgan, 1979). In the present experiment, however we used motor program theory (Schmidt, 1975, 1985) as a theoretical backdrop to aid the a priori determination of the GMP directing the production of three separate three-segment key pressing sequences. Moreover, we used tasks that had different relative timing structures *but shared* the same overall duration, so we could localize any interference generated during acquisition at the level of the GMP.

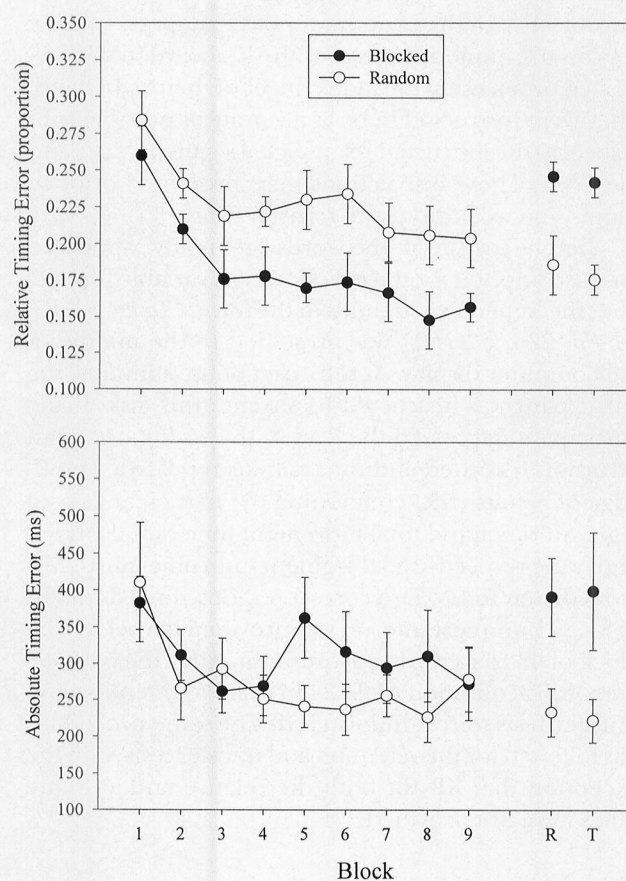


Figure 1. Mean relative timing error (top panel) and absolute timing error (top panel) blocked and random practice acquisition conditions for the acquisition, retention, and transfer blocks (R = retention; T = transfer).

This latter aspect was not present in previous studies addressing this issue (see Sekiya et al., 1994).

The results indicated that blocked practice enhanced performance relative to random practice during acquisition with respect to producing the relative characteristics of the movement (i.e., lower RTE) but not for the scaling of absolute time (i.e., E). These data are consistent with the general finding of a great many studies revealing immediate acquisition benefits from blocked rather than random practice (see Brady, 1998; Magill & Hall, 1990). Also congruent with the general tenet of the CI effect, random practice facilitated superior learning of the practiced tasks as indicated by superior performance during both retention and transfer tests. Importantly, with respect to Magill and Hall's (1990) claim, the data are straightforward. Practicing tasks that differ only with respect to their relative timing *can* provide sufficient interference during training to reap the long-term benefits often associated with random as opposed to blocked practice. Thus, it appears that variability across *either* the invariant (i.e., relative time as in the present experiment) or variant (e.g., absolute time as in Lai & Shea, 1998) timing requirements of a task can support operations engendered by random practice that result in superior learning.

The intriguing aspect of the present data is how the benefit from random practice of movements involving different relative timing structures was manifest. Specifically, in both retention and transfer tests, the benefits were broad in scope. That is, less error in *both* relative and absolute timing was observed for random practice participants compared to their blocked practice counterparts. In the case of the GMP component, the first question one might ask is what was different in this study compared to that of Sekiya et al. (1994) such that GMP improvements were evident only in the present study. The most obvious answer would seem to center on the concomitant shifts in the relative and absolute time structure required by Sekiya et al. (1994) but not in the present study and the associated assumption that such extensive trial-to-trial modifications outweighed any benefit for GMP learning that might typically be afforded from random presentation. Indeed, this is not the first time that a "processing overload" has been identified as a possible source of limitation with respect to the resultant learning that can be garnered from random practice formats (see Wright, Li, & Whitacre, 1992).

If the aforementioned account of GMP learning in the present study is right, one is left to ponder why both the Sekiya et al. (1994) and the present work reveal sizeable improvements in absolute time specification in these diverse "demand" situations. In fact, the superior parameter specification as a function of random practice in the present study was somewhat surprising, because all participants experienced constant practice on

this dimension of the task. These data suggest that as long as parameter specification, in this case scaling absolute time, occurs in the context of random presentation of "some" dimension of the task, benefits can accrue at the level of scaling the absolute timing component. This contention is consistent with that observed in many of the earlier studies in which "schedule randomness" was manipulated with respect to a task's spatial layout, and time scaling (i.e., movement time) was typically superior following random schedules (Lee & Magill, 1983, 1985; Shea & Morgan, 1979).

One final issue considered in the present work was whether some generalizability could be exhibited following random and blocked training to a new timing coordination requirement that is characteristic of a movement that has a new relative time structure (i.e., the transfer case in the present experiment) from that practice during acquisition. The present data indicated that superior transfer, as a result of random practice, can indeed extend beyond parameter specification and can impact the production of the relative time requirement. This finding is important, because, to date, most prior studies only assessed movement generalizability at the level of movement scaling, typically absolute time or force. While there are some limited data available from the early work showing transfer across unique tasks that differed in spatial layout (and maybe unknown to the experimenters also in terms of timing; Shea & Morgan, 1979), this is the first systematic demonstration of more far-reaching transfer at the level of relative time or its theoretical counterpart, the GMP.

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Note

1. The notion of the generalized motor program (GMP) has its roots in schema or motor program theory (Schmidt, 1975, 1985). According to this perspective, the GMP is developed over practice and becomes the structure that governs a class of actions that share the same features (e.g., sequencing, relative timing, relative force, spatial configuration). Further, Schmidt proposed that individual movements governed by the same GMP are produced by the trial-to-trial assignment of movement parameters via the recall schema (e.g., absolute time, absolute force, or the size of the movement).

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